

Midterm 1 (practice version - NOT GRADED), Math 170a, Fall 2018
Instructor: Elizaveta Rebrova

Printed name: _____

Signed name: _____

Student ID number: _____

Instructions:

- Read problems very carefully. If you have any questions please ask.
- The correct final answer alone is not sufficient for full credit - you should explain your answers as much as you can, saving enough time to attempt all the problems. Exception: Problem 1 (see instruction in the beginning of the statement).
- Your final answers do not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- You are allowed to use only the items necessary for writing. Notes, books, sheets, calculators, phones are not allowed, please put them away. If you need more paper please raise your hand.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
Total:	50	

1. For this problem only: please provide short answers (or short explanations when appropriate).

(a) (2 points) If events A and B satisfy $\mathbf{P}(A) = 0.5$ and $\mathbf{P}(A \cap B^c) = 0.2$, find the conditional probability $\mathbf{P}(B|A)$.

(b) (2 points) If A is an event with a strictly positive probability what is $\mathbf{P}(A|A)$?

(c) (2 points) If $\Omega = A \cup B$, $\mathbf{P}(A \cap B^c) = 0.6$, $\mathbf{P}(A^c \cap B) = 0.2$ find the probabilities of A and B .

(d) (2 points) Suppose you are invited to 9 parties and you choose to come to each of them independently with probability 0.3. What is the chance you visit no more than 3 parties?

(e) (2 points) Roll a fair die (6 sides, no bias) and look at the number that comes up. Consider the events $A = \{\text{the number is even}\}$, $B = \{\text{the number is less or equal than 4}\}$ and $C = \{\text{the number is less or equal to 5}\}$. Show that events A and B are independent, but A and C are not independent.

2. (10 points) There are n different cars labeled with numbers 1 to n . These cars are parked in n consecutive parking spaces from left to right in a uniformly random order. What is the probability that the cars labeled 1 and 2 are parked next to each other (meaning that the car 2 is directly to the left or to the right of the car 1)? Simplify your answer.

3. (10 points) A certain department has 20 employees, 10 are instructors and 10 are TAs. This department participates in a lottery in which the probability of winning the 1st prize is 0.1, winning the 2nd prize is 0.2 and winning the 3rd prize is 0.3 (and winning no prize is 0.4). The 1st prize is a dinner of 15 people, the 2nd prize is a dinner for 10 people and the 3rd prize is a dinner for 5 people. If the department wins a prize, the people going to prize dinner will be chosen uniformly at random out of all 20 employees. What is the probability that exactly one instructor from this department goes to the prize dinner?

4. (10 points) A burglar just stole your keychain which has n keys, exactly two of which open your apartment door. He will try to open your door using the keys one by one in a random order. Whenever a key does not open your door, he will not attempt to use this key again. What is the probability that the burglar opens your door on the k th attempt?

5. (10 points) There are three museums in a town. Each is open with probability 0.4 and closed with probability 0.6 independently of the other museums. Two of the museums are located on the street A and one on the street B. A tourist chooses either street A or street B with equal probability and walks down the whole length of the street he chose. He will enter every **open** museum he encounters on his way. Given that he entered exactly one museum, what is the probability he chose to walk down the street A?